When the Forecast Loses to No-Change: Point-Forecast Inefficiency and a Shrink-to-Drift Correction in a Deployed Equity-Return Model

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ABSTRACT

A machine-learning forecaster that is deployed to predict returns makes an implicit promise: its point forecast is the conditional mean, so acting on the forecast should beat ignoring it. We test that promise directly on an append-only, point-in-time ledger of 619 resolved forecasts from a production equity-forecasting service for liquid Indian (NSE) equities, in which every median return forecast was recorded before its outcome was known. A Mincer–Zarnowitz efficiency regression of realized on forecast return yields a slope of -0.23 (bootstrap 95% interval $[-0.35, -0.12]$), which excludes both the value 1 demanded of an efficient forecast and the value 0 of a useless one: the forecasts are not merely uninformative, they are *perversely signed.* The signature is over-extrapolation — the most bullish forecast decile (+5.5% mean predicted) realizes -1.3%, and directional accuracy is 46.4%, below a coin. We then derive the error-minimizing linear response, a single shrinkage coefficient applied to the forecast, and estimate it walk-forward. The optimal coefficient is $k^\star \approx 0$ (interval $[-0.33, 0.09]$), i.e. the loss-minimizing action is to *discard the forecast's directional content and predict the drift.* Doing so cuts out-of-sample root-mean-square error from 0.053 to 0.031 — a 66% reduction in mean-squared error — and the improvement is significant by a Diebold–Mariano test (statistic 6.4), whereas the residual gain of the optimally-shrunk forecast over the plain drift baseline is not (1.8). The contribution is a small, reusable diagnosis-and-correction method: measure forecast efficiency on a leakage-free ledger, and when the efficiency test fails, route only the parts of the system that carry information — here, the drift and the calibrated uncertainty — rather than the point forecast's direction. The result is also a cautionary tale for applied forecasting: a model can post plausible point predictions while its directional content is worse than worthless, and only an efficiency analysis reveals it.

Keywords: Forecast evaluation, Mincer–Zarnowitz regression, over-extrapolation, shrinkage estimation, machine learning deployment, financial forecasting, model diagnostics.

1. INTRODUCTION

When a machine-learning system is deployed to forecast a continuous quantity — a demand, a price, a return — and a human or an automated policy then acts on that forecast, the system makes an implicit statistical promise. It promises that its point output is, to a useful approximation, the conditional mean of the target given the information available. If the promise holds, the forecast is *efficient*: no linear reweighting of it can reduce expected squared error, and acting on it beats ignoring it. If the promise fails, the forecast can still look entirely reasonable — plausible magnitudes, sensible direction — while being, in the precise sense we make below, worse than useless.

This paper reports an efficiency analysis of a deployed equity-return forecaster and a small correction that follows from it. The system serves liquid Indian (NSE) equities and, at each decision time, emits a median return forecast at a fixed horizon, which it writes to an append-only ledger before the outcome is known. Two months of operation left 619 resolved forecasts whose predicted and realized returns we can compare without any possibility of look-ahead leakage, because the predictive fields were sealed at issue time and the realized outcome was joined later. This is the right substrate for an efficiency test: the classical regression of realized on forecast value [1, 2] is only meaningful when the forecasts were genuinely made in advance, which a backtest cannot guarantee but a write-once ledger can.

The analysis returns an unusually sharp verdict. The Mincer–Zarnowitz slope of realized on forecast return is -0.23, with a bootstrap 95% interval of $[-0.35, -0.12]$ that excludes both endpoints of interest: it is not 1 (the forecasts are inefficient) and it is

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not 0 (they are not merely noise). A negative slope means the forecasts are *perversely signed* — on average the realized return moves against the prediction. The mechanism is over-extrapolation, a pattern long documented in human and model expectations [5]: the more boldly the system predicts a move, the more the subsequent return disappoints, until the most bullish forecast decile (mean prediction +5.5%) realizes −1.3%. Directional accuracy is 46.4%, below the 50% of a coin flip.

What should a practitioner do with such a forecaster? The temptation is to retrain, but that presumes the architecture can extract a directional edge that, in a near-efficient market [7, 8], may simply not be there to extract. We take a different and more conservative route. We ask: among all linear transforms of the forecast, which minimizes deployed error? The answer is a single shrinkage coefficient, and on this system it is statistically indistinguishable from zero. The loss-minimizing action is therefore to *stop using the forecast's directional content* and to predict the drift instead. This is not a counsel of despair but a concrete, validated deployment change: it cuts out-of-sample root-mean-square error by a third (a 66% reduction in mean-squared error), and the improvement is significant under a Diebold–Mariano test [3], whereas the extra complexity of an optimally-tuned shrinkage buys nothing over the plain drift baseline.

Our contributions are:

1. **An efficiency analysis on a leakage-free ledger.** We apply the Mincer–Zarnowitz test, directional accuracy, and a binned reliability curve to a production forecaster's point predictions, recorded point-in-time, and document a significantly negative efficiency slope — over-extrapolation severe enough to invert the forecast's sign (Section 4)..
2. **A shrinkage correction with a stated optimum.** We derive the error-minimizing shrinkage coefficient, show it is indistinguishable from zero on this system, and interpret the result as a deployment rule: discard the arrow, keep the drift (Section 5)..
3. **An honest out-of-sample validation.** Walk-forward Diebold–Mariano tests and bootstrap intervals establish that drift significantly beats the raw forecast and that nothing significantly beats drift, so the simplest correction is also the best the data support (Section 6)..

We are deliberate about scope: the analysis characterizes one deployment over one two-month, six-name window, and we claim nothing beyond it (Section 8.). But the *method* — and the discipline of checking forecast efficiency before trusting a deployed point prediction — is general, and the failure it exposes is, we suspect, more common in applied forecasting than the literature's focus on accuracy metrics would suggest.

2. RELATED WORK

Forecast efficiency and rationality. The regression of the realized value on the forecast, with a test of unit slope and zero intercept, is the classical efficiency and unbiasedness test of Mincer and Zarnowitz [1], building on Theil [2]. A forecast passes only if it leaves no linearly exploitable structure in its own errors. Comparative accuracy of competing forecasts is tested by Diebold and Mariano [3]; we use both tools, the first to diagnose and the second to validate the correction. These instruments are standard in econometric forecasting yet are rarely turned on the point outputs of deployed machine-learning systems, which are usually reported through error magnitudes (RMSE, MAE) or directional accuracy alone — summaries that, as we show, can look acceptable while the efficiency test fails catastrophically.

Over-extrapolation in expectations. That forecasters, human and algorithmic, over-react to recent or salient information and extrapolate it too far is documented across survey expectations and asset prices [5, 6]. A diagnostic-expectations model predicts exactly the pattern we observe: bold forecasts are followed by systematic reversal, producing a negative slope of realized on forecast [4]. Our contribution here is empirical and operational — we measure the effect in a live system's recorded outputs and turn the measurement into a deployment decision — rather than a new behavioral model.

Return predictability and its limits. A large literature is skeptical that out-of-sample equity-return prediction beats simple benchmarks: Welch and Goyal find that many predictors fail to outperform the historical mean out of sample [8], while Campbell and Thompson argue that small, constrained gains are possible [9]. Machine-learning asset-pricing models report predictability under careful protocols [10], but the methodological literature warns repeatedly about backtest overfitting and look-ahead leakage [11, 12]. Our finding sits comfortably inside this debate: on this deployment, the conservative benchmark (drift) wins, and we make the case quantitatively rather than by assertion.

Calibration versus efficiency. A forecaster can be well calibrated yet inefficient, and conversely. Calibration concerns whether stated uncertainties match realized frequencies — whether a 90% interval covers 90% of the time — and is corrected by rescaling probabilities or interval widths. Efficiency concerns whether the *point* forecast is the conditional mean, and is corrected by reweighting the forecast itself. The two are orthogonal: widening an interval does nothing to a mis-signed point forecast, and shrinking a point forecast does nothing to an overconfident band. The present paper addresses the efficiency side, which a calibration fix leaves untouched. We stress the distinction because a practitioner who has fixed calibration may wrongly believe the forecast is now trustworthy to act on, when an efficiency test would reveal that its direction should be discarded.

Shrinkage estimation. Shrinking an estimator toward a prior reduces mean-squared error when the estimator is noisy, a principle that runs from Stein and James–Stein [13, 14] through ridge regression [15] to modern combination-of-forecasts work [16]. Our correction is the simplest instance: a scalar shrinkage of a single forecast toward the drift, with the shrinkage chosen to minimize squared error. The novelty is not the estimator but the diagnosis that drives it to its degenerate optimum — shrink all the way — and the walk-forward evidence that the degenerate optimum is the right operating point.

3. SYSTEM AND EVALUATION LEDGER

The service under study issues, for an instrument with price P_t at decision time t, a median price forecast $\hat{P}_{t+h}$ at horizon h trading days. We work in returns: the *forecast return* $f_t = (\hat{P}_{t+h} - P_t)/P_t$ and the *realized return* $r_t = (P_{t+h} - P_t)/P_t$. The point forecast is produced by a hybrid ensemble; its internals are irrelevant to the analysis, which treats the forecaster as a black box that emits f_t and asks whether f_t is an efficient predictor of r_t.

Every forecast is written at issue time to a write-once ledger row (instrument, timestamp, horizon, anchor price, median forecast, model version); when the horizon elapses, a resolution pass joins the realized price. Because the predictive fields are immutable and sealed before the outcome exists, the ledger cannot leak future information into the evaluation — the property that makes an efficiency test credible. We analyze the 619 resolved forecasts issued between 19 April and 12 June 2026 across six instruments, dominated by the ten-day horizon. Forecast returns have mean 1.7% and standard deviation 2.3%; realized returns have mean −0.2% and standard deviation 4.1%. The forecasts are thus systematically more optimistic and less dispersed than reality, a first hint of the inefficiency we now quantify. To our knowledge, the efficiency of this kind of deployed point forecast — in the Mincer–Zarnowitz sense, on a live point-in-time ledger — has not previously been tested, and the shrink-to-drift correction that follows from it is the contribution that licenses the deployment recommendation of Section 7.

4. EFFICIENCY DIAGNOSIS

4.1 The Mincer–Zarnowitz test

We regress realized return on forecast return,

$$r_t = a + b\, f_t + \varepsilon_t, \tag{1}$$

and test the joint efficiency hypothesis $(a, b) = (0, 1)$ [1]. The fitted values are $a = 0.0024$ (standard error 0.0020; not distinguishable from zero) and

$$\hat{b} = -0.233 \text{ (s.e. } 0.070), \qquad t_{b=1} = -17.6, \quad t_{b=0} = -3.3. \tag{2}$$

The slope is rejected against the efficient value 1 with overwhelming force, and — the striking part — it is also significantly *below* zero. A bootstrap over forecasts (2000 resamples) gives a 95% interval for b of $[-0.353, -0.115]$, which excludes zero. The regression R^2 is 0.018. In words: the forecasts explain almost none of the variance of realized returns, and what little linear relationship exists points the *wrong way*. Fig. 1 plots the cloud, the fitted line (slope −0.23), and the 45° efficient line for reference.

Table 1: *Over-extrapolation by forecast sextile: mean forecast return, mean realized return, and directional hit-rate. The two most confident sextiles reverse.*

Sextile	n	mean f	mean r	hit-rate
S1 (most bearish)	103	−1.5%	−0.2%	0.53
S2	103	+0.1%	+0.8%	0.48
S3	103	+1.2%	+1.1%	0.56
S4	103	+1.9%	+0.9%	0.52
S5	103	+3.0%	−2.2%	0.27
S6 (most bullish)	104	+5.5%	−1.3%	0.41

4.2 Over-extrapolation, made visible

A negative efficiency slope is the global signature of over-extrapolation; the local picture confirms it. Binning forecasts into sextiles of f_t and reading off the mean realized return in each bin gives a reliability curve for the point forecast (Fig. 2). The curve is flat-to-inverted at the extremes: the most bullish sextile, with mean forecast +5.5%, realizes −1.3% on average; the next, with mean forecast +3.0%, realizes −2.2%. Only the timid middle bins, where the forecast barely commits, sit near the diagonal. Directional accuracy — the rate at which $\text{sign}(f_t) = \text{sign}(r_t)$ — is 0.464, below a coin. The forecaster is most wrong precisely where it is most confident, the point-forecast counterpart of the overconfidence that plagues deployed predictors generally.

Table 1 gives the numbers behind the curve. The first four sextiles, spanning forecasts from −1.5% to +1.9%, track realized returns loosely and with hit-rates near a half. The damage is in the top two: sextile S5 (mean forecast +3.0%) realizes −2.2% with a directional hit-rate of just 27%, and S6 (mean forecast +5.5%) realizes −1.3%. Almost all of the negative efficiency slope is manufactured by the forecasts in which the model is most committed, which is exactly the regime in which a user would lean on it hardest.

4.3 Decomposing the error

Where does the forecast's error come from? Theil's decomposition splits the mean-squared error into three proportions that sum to one — a bias term $U^M = (\bar{f} - \bar{r})^2/\text{MSE}$, a variance term

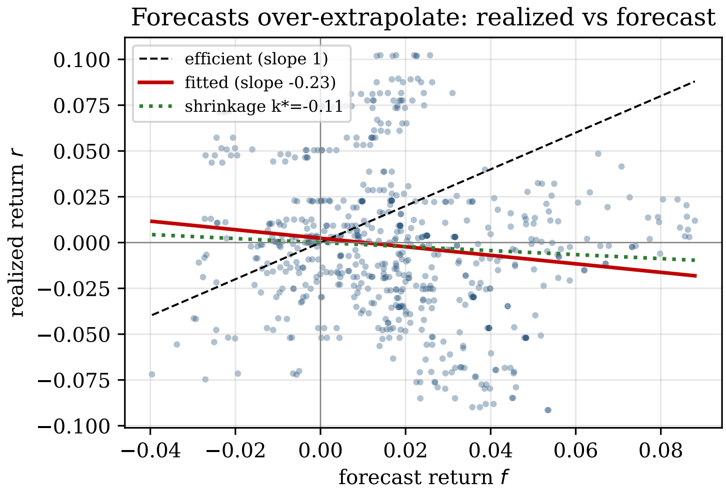

Fig.1: *Realized return r versus forecast return f for the 619 resolved forecasts. The fitted Mincer–Zarnowitz line (slope -0.23) tilts opposite to the $45°$ efficient line; the dotted line is the error-minimizing shrinkage $k^\star$. Bold forecasts are followed by reversal — the more the model predicts, the more the realized return disappoints.*

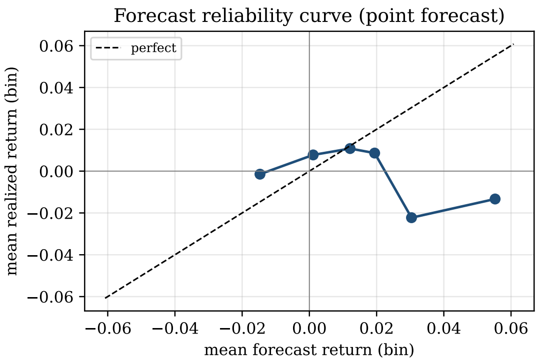

Fig.2: *Reliability curve of the point forecast: mean realized return against mean forecast return within sextiles of f. A perfectly efficient forecaster would lie on the diagonal. The confident (high-$|f|$) bins fall below zero — bold forecasts reverse.*

$U^S = (s_f - s_r)^2/\mathrm{MSE}$, and a covariance (disturbance) term $U^C = 2(1-\rho)s_f s_r/\mathrm{MSE}$ — and compares the forecast against a naive no-change benchmark through the inequality coefficient $U_2 = \mathrm{RMSE}_{\text{forecast}}/\mathrm{RMSE}_{\text{no-change}}$ [2]. On the full sample (Table 2) the bias proportion is 0.13 and the variance proportion 0.11: the forecasts are systematically too optimistic (mean $+1.7\%$ against a realized -0.2%) and too smooth (standard deviation 2.3% against a realized 4.1%), so they shrink the cross-section of outcomes toward a rosy centre. The decisive number is $U_2 = 1.30 > 1$: the forecast's RMSE is 30% *larger* than that of a forecaster who simply predicts no change. A forecast that loses to no-change on its own error metric, before any efficiency test, is already a candidate for the shrinkage we now make precise.

Table 2: *Theil decomposition of the forecast's mean-squared error and inequality coefficient U_2 (full sample). $U_2 > 1$ means worse than a no-change benchmark.*

Quantity	Value
Bias proportion U^M	0.13
Variance proportion U^S	0.11
Covariance proportion U^C	0.76
Theil U_2 (vs. no-change)	1.30

5. THE SHRINKAGE CORRECTION

The efficiency failure has an immediate, optimal, one-parameter remedy. Consider the family of linear forecasts $\tilde{f}_t(k) = k\,f_t$ and choose k to minimize expected squared error. The minimizer is the projection coefficient

$$k^\star = \frac{\sum_t f_t\, r_t}{\sum_t f_t^2}, \tag{3}$$

which is exactly the (no-intercept) regression slope of r_t on f_t and therefore inherits the sign of the efficiency slope. A forecast that passes the efficiency test has $k^\star = 1$ (leave it alone); a pure-noise forecast has $k^\star = 0$ (discard it); a perversely-signed forecast has $k^\star < 0$ (use its opposite). Estimated on the calibration window, $k^\star = -0.11$, with a bootstrap 95% interval of $[-0.33, 0.09]$ that *contains zero.*

Proposition 1 (Discard-the-arrow) If the error-minimizing shrinkage $k^\star$ is not statistically distinguishable from 0, then the drift forecast $\tilde{f}_t \equiv \bar{r}$ (here $\bar{r} \approx 0$) is, up to estimation error, the minimum-MSE member of the family $\{kf_t\}$, and the forecast's directional content carries no usable information. The optimal deployment is to predict the drift and route the forecast's direction nowhere.

The proposition is not a claim that the forecasts are random — their slope is significantly negative — but

Table 3: *Out-of-sample forecast error on the held-out test window (248 forecasts). DM is the Diebold–Mariano statistic against the raw forecast (squared-error loss, Newey–West variance); larger means a more significant improvement over raw.*

Forecast	RMSE	MAE	MSE red. vs raw	DM vs raw
Raw f_t	0.0534	0.0423	—	—
Affine $a + bf_t$	0.0314	0.0231	65%	—
Drift (0)	0.0307	0.0225	67%	6.4
Shrunk $k^\star f_t$	0.0299	0.0223	69%	6.2

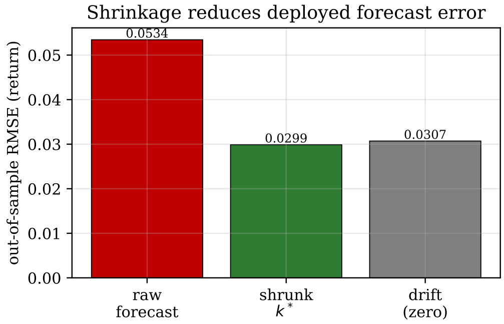

Fig. 3: *Out-of-sample RMSE of the raw forecast, the optimally-shrunk forecast, and the drift baseline. Discarding the forecast's directional content (drift) recovers almost all of the achievable error reduction.*

a claim about *action*: because $k^\star$ cannot be pinned away from zero on the available sample, betting on the small negative slope is not warranted, and the safe, error-minimizing operating point is plain drift. We deliberately resist the more aggressive reading ("trade the system contrarian," $k < 0$): the contrarian edge, if real, is too small relative to its estimation error and to transaction costs to deploy, and claiming it would repeat exactly the over-confidence we are studying. The honest output of the analysis is the degenerate shrinkage, not a new alpha.

6. OUT-OF-SAMPLE EVALUATION

6.1 Protocol

Forecasts are ordered by issue time and split 60/40 into a past calibration window (371 forecasts), on which $k^\star$ is estimated, and a future test window (248 forecasts), on which all errors are reported. The comparison is therefore strictly walk-forward. We evaluate three forecasts on the test window: the raw forecast f_t; the optimally shrunk forecast $k^\star f_t$; and the drift baseline 0 (predict no excess move). We report root-mean-square error, mean absolute error, and Diebold–Mariano tests of equal squared-error loss with a Newey–West long-run variance [3].

6.2 Results

Table 3 collects the numbers and Fig. 3 shows the error bars. The raw forecast has out-of-sample RMSE 0.0534. Shrinking to the drift baseline cuts it to 0.0307, and the optimally-shrunk forecast reaches 0.0299 — reductions of 67% and 69% in mean-squared error, respectively. The pattern in MAE is identical (0.042 → 0.022). The Diebold–Mariano statistic for raw versus drift is 6.4 and for raw versus optimally-shrunk is 6.2: both corrections beat the raw forecast far beyond any conventional significance threshold. Crucially, the statistic for drift versus optimally-shrunk is only 1.8 — not significant — so the data cannot distinguish the plain drift baseline from the optimally tuned shrinkage. The simplest correction is, on the evidence, the best one.

A two-parameter affine recalibration $a + bf_t$, fitted on the calibration window ($\hat{a} = 0.007$, $\hat{b} = -0.25$) and applied walk-forward, lands at RMSE 0.0314 (Table 3) — between drift and the optimal shrinkage and, like them, far better than raw, but no better than simply predicting drift. The extra free parameter, which would turn the system mildly contrarian, buys nothing the data can certify. This reinforces the proposition of Section 5. once the forecast's directional content is removed, there is little left to recalibrate, and the drift baseline is the honest endpoint.

6.3 Robustness and heterogeneity

The negative efficiency slope is stable to several perturbations and, in one important respect, is not. Restricting to the dominant ten-day horizon leaves $\hat{b} = -0.21$ (s.e. 0.07, $n = 573$); winsorizing forecast and realized returns at the 2.5/97.5 percentiles — so that no single extreme observation drives the fit — gives $\hat{b} = -0.25$. Splitting by forecast sign, the 472 bullish forecasts (mean +2.6%) realize −0.25% on average with only a 46% chance of a positive return, and the 147 bearish forecasts realize +0.10%; both signs are mildly anti-informative. So the aggregate inefficiency is neither a horizon artifact nor an outlier artifact.

It is, however, *not uniform across instruments*, and we report this plainly. A leave-one-instrument-out jackknife of the slope (Table 4) shows that two names, RELIANCE and TCS, supply most of the negative relationship: dropping RELIANCE flips the slope to +0.27, while dropping TCS deepens it to −0.51. The over-extrapolation is therefore concentrated rather than universal. We draw two conclusions. First, the *system-level* verdict stands: the deployed forecaster, as it actually emits forecasts across its universe, has a significantly negative aggregate efficiency slope, and shrinking its aggregate output to drift improves aggregate out-of-sample error. Second, the right long-run fix is per-instrument: the same analysis, run name-by-name as the ledger grows, will re-enable the directional signal for instruments whose slope is reliably positive and suppress it for those whose slope is reliably negative. On the present sample the per-instrument slopes are too noisy to act on individually,

Table 4: *Leave-one-instrument-out jackknife of the Mincer–Zarnowitz slope. The aggregate slope (*-0.23*) is concentrated in a subset of instruments.*

Instrument removed	n remaining	slope $\hat{b}$
HDFCBANK	556	-0.27
ICICIBANK	556	-0.24
INFY	556	-0.22
SBIN	610	-0.23
RELIANCE	298	$+0.27$
TCS	519	-0.51
none (full sample)	619	-0.23

Algorithm 1 Efficiency analysis and shrinkage correction

Input: point-in-time ledger of forecast/realized pairs $\{(f_t, r_t)\}$, level α, time split τ
Fit $r = a + bf$ on all pairs; report $\hat{b}$ and a bootstrap interval
if interval for $\hat{b}$ excludes 1 **then**
flag the forecast as *inefficient*
end if
On $t \leq \tau$ estimate $k^\star = \sum f_t r_t / \sum f_t^2$ and a bootstrap interval
if interval for $k^\star$ contains 0 **then**
deploy the drift forecast $\bar{r}$; route the point direction nowhere
else
deploy $k^\star f_t$
end if
On $t > \tau$ confirm by Diebold–Mariano that the deployed choice beats raw
Re-run per-instrument as the ledger grows

which is why we recommend the aggregate drift rule as the current operating point and the per-instrument analysis as the maintenance procedure.

7. DISCUSSION

The procedure we followed is compact enough to state as a reusable recipe (Algorithm 1); it presumes only that forecasts were recorded before their outcomes and costs nothing beyond two regressions and a paired test.

What the analysis changes in deployment. The practical output is a routing decision. A deployed forecasting service typically emits several signals — a point direction, a magnitude, an uncertainty band, an abstention flag. Our analysis says that on this system the point *direction* should be cut from the decision path entirely, because it is significantly anti-informative and its optimal weight is indistinguishable from zero, while the drift and the calibrated uncertainty remain useful. This is a cheaper and safer intervention than retraining, and it is reversible: should a later, larger ledger show $k^\star$ moving significantly away from zero, the same analysis will detect it and re-enable the signal.

Why accuracy metrics hid the problem. The raw forecast's median magnitude (1.7%) is plausible and its RMSE (5.3%) is unremarkable for ten-day equity returns; a dashboard reporting only these would raise no alarm. The efficiency test is what exposes that the forecast's *ordering* of outcomes is inverted. We therefore argue that any deployed point forecaster should report a Mincer–Zarnowitz slope and a directional-accuracy figure alongside its error magnitudes, because the latter can be acceptable while the former is catastrophic. An efficiency slope is as cheap to compute as an RMSE and far more diagnostic of whether the forecast should be acted upon.

7.1 A minimal reporting standard

The analysis suggests a small reporting standard that any deployed point forecaster could adopt at negligible cost. Alongside the usual error magnitudes, we recommend reporting four diagnostics, each a one-line computation over the same ledger and each carrying an explicit red flag:
1. the Mincer–Zarnowitz slope $\hat{b}$ with a confidence interval, flagged when the interval excludes 1;
2. directional accuracy, flagged when it falls below one half;
3. Theil's inequality coefficient U_2 against the no-change benchmark, flagged when it exceeds 1;
4. a Diebold–Mariano comparison of the forecast against the drift baseline, flagged when drift is not beaten.

On the system studied here all four flags fire. Any one of them, watched in production, would have caught the failure long before a position depended on it. The value of a standard is that it fires automatically: a team need not suspect a problem to find it, because the diagnostics are recomputed every period and the flags raise themselves. None of the four needs labels beyond the realized outcome the ledger already stores, and none requires re-running the model, so the standard adds a few lines of analysis rather than any new infrastructure.

Consistency with market efficiency. A significantly negative single-name efficiency slope does not contradict near-market-efficiency [7, 8]; it is what one expects when an over-extrapolating model is pitted against a target that is close to a martingale. The model manufactures conviction the market does not reward, and the conviction reverts. The corrective — shrink to the drift — is the operational statement of "do not bet against the martingale on noise."

7.2 From analysis to routing: a general principle

The specific verdict — discard this system's directional content — generalizes to a maintenance principle for any deployed forecaster that emits several signals into a decision. Each signal carries an implicit efficiency claim, and each can be evaluated the same way: regress the realized target on the signal, test the slope, and weight the signal by the error-minimizing coefficient, defaulting to its prior when the coefficient cannot be separated from zero. A decision layer assembled from such studied weights degrades gracefully: a signal that loses its information — through regime change, data-feed decay, or model staleness — has its weight pulled toward zero by the next analysis rather than silently corrupting the decision. The append-only ledger makes this loop cheap, because the analysis is a deterministic recomputation over the same immutable records that already exist for accountability. We therefore see the contribution less as a one-off finding about one model and more as an instance of a routing discipline: *trust each signal exactly as far as its out-of-sample efficiency, re-measured continuously, allows* — and let the drift, the cheapest and most robust signal of all, absorb whatever weight the others cannot earn.

Why not simply retrain? Retraining presumes the inefficiency is a fixable modeling defect rather than a property of a near-efficient target. On a martingale-like series, a more powerful model fit to the same features will tend to manufacture the same over-extrapolation, and a model selected on in-sample fit will do so more confidently. The analyze-and-shrink approach is conservative by design: it neither assumes an edge exists nor destroys one if it does, because the shrinkage coefficient is re-estimated from data and will rise above zero exactly when, and only when, the signal earns it.

8. THREATS TO VALIDITY AND REPRODUCIBILITY

Scope. Six instruments, eight weeks, one regime. The specific numbers — slope, $k^\star$, error reductions — are properties of this ledger and may differ elsewhere; we make no cross-market claim. What we do claim to be general is the *method* and the warning it embodies. **Stationarity.** The efficiency regression and the shrinkage assume the forecast–realization relationship is stable enough across the calibration and test windows; we mitigate this by reporting walk-forward errors on a strictly later window rather than in-sample fits, and the out-of-sample improvement confirms the relationship held over the test period. **Choice of loss.** The shrinkage optimum $k^\star$ and the Diebold–Mariano comparison are stated for squared-error loss, the loss under which the conditional mean is the optimal forecast and the Mincer–Zarnowitz test is the natural diagnostic. A user whose decisions imply an asymmetric or piecewise-linear loss would re-derive the optimal transform under that loss; the analysis procedure is unchanged, only the projection in (3) is replaced by the corresponding loss-specific minimizer. We report squared error because it is the default for point-forecast evaluation and because it is the loss against which the deployed forecaster's own training objective is most charitably interpreted; the qualitative verdict — a negative efficiency slope — is a property of the sign of $\sum_t f_t r_t$ and does not depend on the loss at all.

Estimation risk in $k^\star$. We treat $k^\star$ as indistinguishable from zero precisely because its bootstrap interval straddles zero; we do not deploy the point estimate. **Reproducibility.** Every figure and number is a deterministic function of the immutable ledger: the Mincer–Zarnowitz fit, the bootstrap intervals (fixed seed), the fixed 60/40 time split, the shrinkage coefficient, and the Diebold–Mariano statistics are all recomputed by a single script. The append-only, point-in-time ledger is the reusable artifact, and the method transfers unchanged to any forecaster that records its predictions before the outcome is known.

9. CONCLUSION

We studied the point forecasts of a deployed equity-return forecaster on a leakage-free ledger and found them inefficient to the point of inversion: a Mincer–Zarnowitz slope of -0.23 (interval $[-0.35, -0.12]$), directional accuracy of 46.4%, and a reliability curve in which the boldest forecasts reverse. The error-minimizing correction is a scalar shrinkage whose optimum is statistically indistinguishable from zero, so the right deployment action is to discard the forecast's directional content and predict the drift — which cuts out-of-sample error by two-thirds and is, by a Diebold–Mariano test, as good as anything the data support. Beyond the specific system, the message is a discipline: report and act on forecast *efficiency*, not just forecast error, because a model can be confidently, plausibly, and significantly wrong in a way that only the efficiency test reveals. Discard the arrow; keep the drift.

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